

Verona through Rome

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1 Introduction

We present *Roma*^M, a more complete, fully formalized version of *Rome*^M[Nguyen 2026].

2 Model

The configuration remains unchanged from the report. We make one change to the semantics, specifically within `makeRegion`:

$$\begin{array}{c}
 \text{MAKE-REGION} \\
 \frac{
 \begin{array}{l}
 F = (rid, bvar, omap, vmap, fid) \\
 \langle (S :: F), H \rangle.\text{freshObjectId} = oid' \quad \langle (S :: F), H \rangle.\text{freshRegionId} = rid' \\
 R' = (oid', [oid' \mapsto \emptyset], Closed) \quad F' = (rid, bvar, omap, vmap[x \mapsto Rld(rid')], fid)
 \end{array}
 }{
 \langle (S :: F), H \rangle \xrightarrow{\text{makeRegion } x} \langle S :: F', H[rid \mapsto R'] \rangle
 }
 \end{array}$$

In the report, the `regionId` would coincide with the `objectId` of the bridge object, but here we've chosen to completely decouple them. The original reasoning behind this was to future-proof for deallocation, but in writing this we found out that the argument didn't hold, and it would've been fine to leave `makeRegion` as before. Luckily, this doesn't change much, it only means that we need to prove regions are unique separately from objects, as opposed to using a combined argument before.

3 Invariants

We add two new invariants, presented here informally and formally.

Definition 3.1 (HS1). Every reference to an object must be to an object that exists in the configuration.

$$\forall oid : [Old(oid) \in \langle S, H \rangle.\text{refs} \implies oid \in \langle S, H \rangle.\text{objectIds}]$$

Definition 3.2 (HS2). Every reference to a region must be to a region that exists in the configuration.

$$\forall rid : [Rld(rid) \in \langle S, H \rangle.\text{refs} \implies rid \in \text{dom}(H)]$$

These came up in the process of proving *S2* and *S3* for `makeObjStack`, `makeObjRegion`, and `makeRegion`. This stems from the fact that our `freshObjectId` and `freshRegionId` functions only take into account the `Ids` used as keys when creating a new `Id`. Thus, without this invariant, it is technically possible¹ for a configuration to contain a dangling reference to an `objectId` or `regionId` that does not exist within the configuration, but is then inserted via one of the three `make` operations.

¹Although it isn't possible within the context of an operational semantics starting with the starting configuration, which is why we were able to prove it without altering the operational semantics.

4 Reachability

We introduce the notion of reachability. It is the same term as in the original report, but the definition here covers a wider spectrum.

Definition 4.1 (Stack-reachable). An object or region is stack-reachable if it is frame-reachable from some frame in the stack.

$$\frac{\langle S, H \rangle \vdash FReachable(_, a)}{\langle S, H \rangle \vdash SReachable(a)}$$

Definition 4.2 (Frame-reachable). An object or region is frame-reachable if there is a path from a frame to said object or region.

$$\begin{array}{c} \text{VAR} \\ \frac{F = (_, _, _, vmap, fid) \quad vmap(_) = a}{\langle S :: F, H \rangle \vdash FReachable(fid, a)} \end{array} \quad \begin{array}{c} \text{BRIDGE} \\ \frac{F = (rid, _, _, fid) \quad H(rid) = (boid, _, _)}{\langle S :: F, H \rangle \vdash FReachable(fid, Old(boid))} \end{array}$$

$$\frac{\text{STEP} \quad \langle S, H \rangle \vdash FReachable(fid, a) \quad \langle S, H \rangle \vdash a \rightarrow b}{\langle S, H \rangle \vdash FReachable(fid, b)}$$

Definition 4.3 (Region-reachable). An object or region is region-reachable if there is a path from the bridge object to said object or region.

$$\frac{\text{BRIDGE} \quad H(rid) = (boid, _, _)}{\langle S, H \rangle \vdash RReachable(rid, Old(boid))} \quad \frac{\text{STEP} \quad \langle S, H \rangle \vdash RReachable(rid, a) \quad \langle S, H \rangle \vdash a \rightarrow b}{\langle S, H \rangle \vdash RReachable(rid, b)}$$

Definition 4.4 (Reachable-step). A one-step reachable relation between objects and regions. An object or region is one-step reachable from another object if that object contains a reference to said object or region. An object is one-step reachable from a region if the region is closed and the object is the region's bridge object.

$$\frac{\text{OBJ-STEP} \quad Old(oid).objAt(\langle S, H \rangle) = obj \quad b \in obj.refs}{\langle S, H \rangle \vdash Old(oid) \rightarrow b} \quad \frac{\text{REGION-STEP} \quad H(rid) = (boid, _, Closed)}{\langle S, H \rangle \vdash Rld(rid) \rightarrow Old(boid)}$$

THEOREM 4.5. *Region-reachability is the transitive closure of the reachable-step from the region's bridge object.*

$$\langle S, H \rangle \vdash RReachable(rid, a) \iff \exists boid : [H(rid) = (boid, _, _) \wedge Old(boid) \xrightarrow{*} a]$$

THEOREM 4.6. *Frame-reachability is the transitive closure of the reachable-step from the frame.*

$$\langle S, H \rangle \vdash FReachable(fid, a) \iff \exists start : [\langle S, H \rangle \vdash FRoot(fid, start) \wedge start \xrightarrow{*} a]$$

Definition 4.7 (Frame-root). The root coming from a frame can be a reference stored by a variable or the bridge object of that frame's region.

$$\frac{\text{VAR} \quad F = (_, _, _, vmap, fid) \quad vmap(_) = a}{\langle S :: F, H \rangle \vdash FRoot(fid, a)} \quad \frac{\text{BRIDGE} \quad F = (rid, _, _, fid) \quad H(rid) = (boid, _, _)}{\langle S :: F, H \rangle \vdash FRoot(fid, Old(boid))}$$

5 Related Work

6 Conclusion

A Configuration

$RegionId, ObjectId, Index \subseteq Int$
 $VarName, FieldName \subseteq String$

$RuntimeConfiguration \triangleq Heap \times Stack$

$Heap \triangleq RegionId \rightarrow Region$
 $Region \triangleq ObjectId \times ObjMap \times Status$
 $Status \triangleq Open \mid Closed$

$Stack \triangleq List\ Frame$
 $Frame \triangleq RegionId \times VarName \times ObjMap \times VarMap \times Index$

$ObjMap \triangleq ObjectId \rightarrow FieldMap$

$Object \triangleq FieldMap$
 $FieldMap \triangleq FieldName \rightarrow Reference$
 $VarMap \triangleq VarName \rightarrow Reference$

$Reference \triangleq Rld(RegionId) \mid Old(ObjectId)$

B Operational Semantics

VAR-ASGN-STACK

$$\frac{F = (rid, bvar, omap, vmap, fid) \quad x \neq bvar \quad resolveV(x, \langle (S :: F), H \rangle) = \epsilon \quad resolveFA(y.f, \langle (S :: F), H \rangle) = Old(oid)}{\langle (S :: F), H \rangle \xrightarrow{x := y.f} \langle S :: (rid, bvar, omap, vmap[x \mapsto Old(oid)], fid), H \rangle}$$

VAR-ASGN-BRIDGE

$$\frac{F = (rid, x, _, _, _) \quad resolveFA(y.f, \langle (S :: F), H \rangle) = Old(oid) \quad Old(oid).loc(\langle (S :: F), H \rangle) = Rgn(rid) \quad H(rid) = (boid, omap, status)}{\langle (S :: F), H \rangle \xrightarrow{x := y.f} \langle (S :: F), H[rid \mapsto (oid, omap, status)] \rangle}$$

FIELD-ASGN-STACK

$$\begin{array}{c}
F = (rid, bvar, omap, vmap, fid) \\
\text{resolveV}(x, \langle (S :: F), H \rangle) = \text{Old}(oid) \quad \text{Old}(oid).loc(\langle (S :: F), H \rangle) = \text{Stk}(fid) \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(oid') \quad omap(oid) = obj \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: (rid, bvar, omap[oid \mapsto obj[f \mapsto \text{Old}(oid')]], vmap, fid)), H \rangle
\end{array}$$

FIELD-ASGN-REGION

$$\begin{array}{c}
F = (rid, _, _, _) \quad \text{resolveV}(x, \langle (S :: F), H \rangle) = \text{Old}(oid) \\
\text{Old}(oid).loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(oid') \\
\text{Old}(oid').loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad H(rid) = (boid, omap, Open) \quad omap(oid) = obj \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F), H[rid \mapsto (boid, omap[oid \mapsto obj[f \mapsto \text{Old}(oid')]], Open)] \rangle
\end{array}$$

SWAP-STACK

$$\begin{array}{c}
F = (rid, bvar, omap, vmap, fid) \\
x \neq bvar \quad vmap(x) = xref \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \\
\text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \quad \text{Old}(yoid).loc(\langle (S :: F), H \rangle) = \text{Stk}(fid) \\
omap(yoid) = obj \quad omap' = omap[yoid \mapsto obj[f \mapsto xref]] \\
vmap' = vmap[x \mapsto yfref] \quad F' = (rid, bvar, omap', vmap') \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H \rangle
\end{array}$$

SWAP-REGION-OBJECT

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \\
x \neq bvar \quad vmap(x) = \text{Old}(xoid) \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \\
\text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \quad \{\text{Old}(xoid), \text{Old}(yoid)\}.loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \quad obj' = obj[f \mapsto \text{Old}(xoid)] \\
R' = (boid, romap[yoid \mapsto obj'], Open) \quad F' = (rid, bvar, fomap, vmap[x \mapsto yfref]) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H[rid \mapsto R'] \rangle
\end{array}$$

SWAP-REGION-REGION

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \quad x \neq bvar \quad vmap(x) = \text{Rld}(xrid) \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \\
\text{Old}(yoid).loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad H(xrid) = (_, _, \text{Closed}) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \quad obj' = obj[f \mapsto \text{Rld}(xrid)] \\
R' = (boid, romap[yoid \mapsto obj'], Open) \quad F' = (rid, bvar, fomap, vmap[x \mapsto yfref]) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H[rid \mapsto R'] \rangle
\end{array}$$

SWAP-REGION-BRIDGE

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \quad x = bvar \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = \text{Old}(yfoid) \\
\{\text{Old}(yoid), \text{Old}(yfoid)\}.loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \\
obj' = obj[f \mapsto \text{Old}(boid)] \quad R' = (yfoid, romap[yoid \mapsto obj'], Open) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F), H[rid \mapsto R'] \rangle
\end{array}$$

$$\begin{array}{c}
\text{ENTER} \\
\frac{F = (rid, bvar, fomap, vmap, fid) \quad \text{resolveFA}(x.f, \langle S :: F, H \rangle) = \text{Rld}(rid') \quad H(rid') = (boid, romap, Closed) \quad F' = (rid', a, \emptyset, fid + 1)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{enter } x.f \ a} \langle \langle S :: F :: F' \rangle, H[rid' \mapsto (boid, romap, Open)] \rangle} \\
\\
\text{EXIT} \\
\frac{F = (rid, _, _, _, _) \quad H(rid) = (boid, omap, Open)}{\langle \langle S :: F' :: F \rangle, H \rangle \xrightarrow{\text{exit}} \langle S :: F', H[rid \mapsto (boid, omap, Closed)] \rangle} \\
\\
\text{MAKE-OBJ-STACK} \\
\frac{F = (rid, bvar, omap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeObjStack } x} \langle S :: (rid, bvar, omap[oid \mapsto \emptyset], vmap[x \mapsto \text{Old}(oid)], fid), H \rangle} \\
\\
\text{MAKE-OBJ-REGION} \\
\frac{F = (rid, bvar, fomap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid \quad H(rid) = (boid, romap, Open) \quad R = (boid, romap[oid \mapsto \emptyset], Open)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeObjRegion } x} \langle S :: (rid, bvar, fomap, vmap[x \mapsto \text{Old}(oid)], fid), H[rid \mapsto R] \rangle} \\
\\
\text{MAKE-REGION} \\
\frac{F = (rid, bvar, omap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid' \quad \langle \langle S :: F \rangle, H \rangle.freshRegionId = rid' \quad R' = (oid', [oid' \mapsto \emptyset], Closed) \quad F' = (rid, bvar, omap, vmap[x \mapsto \text{Rld}(rid')], fid)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeRegion } x} \langle S :: F', H[rid \mapsto R'] \rangle} \\
\\
\text{MERGE} \\
\frac{F = (rid, bvar, omap, vmap, fid) \quad vmap(x) = \text{Rld}(rid') \quad H(rid) = (boid, romap, Open) \quad H(rid') = (boid', romap', Closed) \quad F' = (rid, bvar, omap, vmap[x \mapsto boid'], fid) \quad R = (boid, romap \cup romap', Open)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{merge } x} \langle S :: F', H[rid' \mapsto \epsilon][rid \mapsto R] \rangle}
\end{array}$$

Definition B.1 (Name). A configuration σ satisfies P if

$$\forall x. \quad |\{y \mid y \in X(\sigma), y = x\}| \leq 1.$$

Definition B.2 (Name).

$$\forall x \in X(\sigma). \exists y. P(x, y) \wedge Q(y).$$

Definition B.3 (Step relation).

$$\frac{P(\sigma, a) \quad Q(a, f, b)}{\sigma \vdash a \xrightarrow{f} b}$$

Definition B.4 (Reachable).

$$\text{Reachable}(\sigma, i, b) \iff \exists a. \text{Root}(\sigma, i, a) \wedge \sigma \vdash a \xrightarrow{*} b.$$

Definition B.5 (Two-branch step relation).

$$\begin{array}{c}
\text{STEP-OBJECT} \\
\frac{P(\sigma, a) = c \quad c.f = b}{\sigma \vdash a \xrightarrow{r} b} \\
\\
\text{STEP-REGION} \\
\frac{Q(\sigma, a) = c \quad \text{guard}(c) \quad b = \text{target}(c)}{\sigma \vdash a \xrightarrow{r} b}
\end{array}$$

Definition B.6 (Inductively-closed reachability).

$$\frac{\text{BASE-A} \quad x \in X(\sigma) \quad P(x) = a}{\text{Reach}(\sigma, i, a)}$$

$$\frac{\text{BASE-B} \quad x \in X(\sigma) \quad Q(x) = a}{\text{Reach}(\sigma, i, a)}$$

$$\frac{\text{STEP} \quad \sigma \vdash a \xrightarrow{f} b \quad \text{Reach}(\sigma, i, a)}{\text{Reach}(\sigma, i, b)}$$

THEOREM B.7 (NAME). *Statement.*

THEOREM B.8 (NAME). *Let σ be such that $P(\sigma)$. Then*

$$Q(\sigma) \implies R(\sigma).$$

References

Hoang Nguyen. 2026. *Verona through Rome - Revisiting Verona's model and region system*. Technical Report. Imperial College London. <https://github.com/hxcuber/verona-through-rome/blob/main/report.pdf>